

of the receiver unit is shown in Fig. 7 and its component layout in Fig. 8.

You can also assemble the circuit on a general-purpose PCB. Before mounting the MCUs on the PCBs, burn the respective codes into the MCUs using a suitable programmer board. After mounting all components on the transmitter and receiver PCBs, switch on the power supplies on the respective units. LCD1 in the transmitter unit will display Welcome (Fig. 9) and LCD2 in the receiver unit will display Waiting 4 Orders (Fig. 10). Presets VR1 and VR2 are used for contrast control of the LCD displays. If there is any display problem, press reset switch momentarily and check the display again. For further troubleshooting the circuit, check voltages at various test points as given in the table.

Ordering procedure

1. Assuming the transmitter unit is installed near the table where the

customer is seated, when he or she presses a key, the menu appears. The user can browse items/menu using Up/Down keys. A typical menu is shown in Fig. 11.

2. If the user wants a particular item, he or she can select that item by pressing Enter, along with the quantity and again press Enter to confirm the order.

3. If the user wants to cancel the order, he or she can press Clear/Cancel.

4. After confirming the order, the item and quantity are sent to the receiver through RF wireless communication. The item and quantity are displayed on the LCD in the receiver unit. The receiver can be installed in the kitchen or in the room where the hotel manager sits.

5. The bill amount is calculated instantly after every order. The user can view the bill and verify at any time by pressing Bill.

6. While the user is waiting, he or she can play the game by pressing Game. The game is shown in Fig. 12.

Future scope

The cost-effectiveness and scope for extensive expansion proves to be effective not only for small and medium restaurants but also for premium and luxury hotels.

1. This technology can be coupled with RFID tags (like ID cards used in colleges or offices) for advance ordering of food items in college and office canteens.

2. Data analysis can be done by connecting to an external data processing machine, for example, a regular personal computer.

3. Extra voice channel transmission for spe-



Fig. 9: Welcome message in the transmitter unit



Fig. 10: Message in the receiver unit



Fig. 11: Typical menu on the LCD



Fig. 12: The game

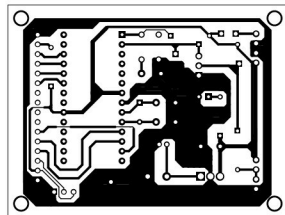


Fig. 7: Actual-size PCB of the receiver unit

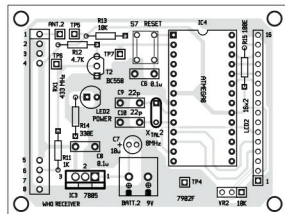


Fig. 8: Component layout of the PCB of the receiver unit

EFY Note

The source codes of this project are included in this month's EFY DVD and are also available for free download at source.efymag.com

cial interaction can be done with the chef.

4. Automated serving system (pneumatic/robotic) coupled with this system can be implemented for a completely automated hotel. ●



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